

CLUB club

2026-09-18

2026-09-18 class

Last week we took some raw FASTQ files and trimmed adapters from them using `fastp`.

Who is actually doing anything on the cluster between classes?

Today we'll learn how to "map" the reads to references and (maybe) do some command-line practice. "Mapping" is also often called "aligning".

- `bowtie2`
- `bwa`

Starting from trimmed FASTQ

Now that you've trimmed adapters from your FASTQ, what do you do next?

You likely want to align them against a reference sequence (or sequences). That's what I think people do in Geneious when they ask us if they can have the raw reads for a sequencing run.

Let's work in a new directory

```
$ mkdir 20260918-map  
$ cd 20260918-map
```

Make a bowtie2 index

```
$ bowtie2-build --help # Have read of this (pipe the output to less).  
$ bowtie2-build ~/data/references/HBV-G.fasta HBV
```

```
# The above command produces the below files.
```

```
$ ls -l  
total 768  
-rw-r--r-- 1 jonestc posix-nogroup 4195559 Sep 17 22:01 HBV.1.bt2  
-rw-r--r-- 1 jonestc posix-nogroup      816 Sep 17 22:01 HBV.2.bt2  
-rw-r--r-- 1 jonestc posix-nogroup      17 Sep 17 22:01 HBV.3.bt2  
-rw-r--r-- 1 jonestc posix-nogroup     812 Sep 17 22:01 HBV.4.bt2  
-rw-r--r-- 1 jonestc posix-nogroup 4195559 Sep 17 22:01 HBV.rev.1.bt2  
-rw-r--r-- 1 jonestc posix-nogroup      816 Sep 17 22:01 HBV.rev.2.bt2
```

Some Hepatitis B virus FASTQ

I put some HBV FASTQ data into the stored area:

```
# Remember, I have a symlink in '~/data' that points to
# /sc-projects/sc-proj-cc11-civclub/club-club/data/fastq

$ ls -la ~/data/fastq/HBV_*
-rwxrwxr--+ 1 jonestc posix-nogroup 29415281 Sep 17 21:14 HBV_RISE718_R1_trimmed.fastq.gz
-rwxrwxr--+ 1 jonestc posix-nogroup 29693520 Sep 17 21:14 HBV_RISE718_R2_trimmed.fastq.gz
-rw-r--r-- 1 jonestc posix-nogroup 228117891 Sep 17 22:32
HBV_RISE718_single_trimmed.fastq.gz
```

The **R1** and **R2** files each have 1,000,000 reads.

How do we see how many reads are in **HBV_RISE718_single_trimmed.fastq.gz**? (answer: 5,494,312)

If you want more data, see

```
$ ls LV7008876184-LV7008414499-CGG2015787_S1_L002_*.fastq.gz
LV7008876184-LV7008414499-CGG2015787_S1_L002_R1_001-ar3.fastq.gz
LV7008876184-LV7008414499-CGG2015787_S1_L002_R2_001-ar3.fastq.gz
```

These files are both 8GB and each contains 282,894,963 reads. I.e. 0.56 billion reads in total.

Do the mapping

```
$ time bowtie2 --local --xeq --no-unal -x HBV \  
-1 ~/data/fastq/HBV_RISE718_R1_trimmed.fastq.gz \  
-2 ~/data/fastq/HBV_RISE718_R2_trimmed.fastq.gz \  
> matches-local-without-single.sam
```

This takes about 8 seconds.

Or if you want a few more matches, include the unpaired reads (using **-U**):

```
$ time bowtie2 --local --xeq --no-unal -x HBV \  
-1 ~/data/fastq/HBV_RISE718_R1_trimmed.fastq.gz \  
-2 ~/data/fastq/HBV_RISE718_R2_trimmed.fastq.gz \  
-U ~/data/fastq/HBV_RISE718_single_trimmed.fastq.gz \  
> matches-local-with-single.sam
```

Which takes about 40 seconds.

About those bowtie2 options

- `--local`: Allow local (i.e., within-read) matches as opposed to end-to-end in which the whole read must match. When you use local matching, read ends can be "soft clipped" (i.e., marked in the CIGAR string as not matching). With the bowtie2 default (end-to-end matching) this is not allowed.
- `--xeq`: Use `X` and `=` in the CIGAR string, instead of just `M`.
- `--no-unal`: Don't output anything for reads that don't align
- `-x HBV`: Specifies the bowtie2 index to map reads against.

SAM files

bowtie2 produces SAM output. See [https://en.wikipedia.org/wiki/SAM_\(file_format\)](https://en.wikipedia.org/wiki/SAM_(file_format))

SAM files are TSV (TAB-separated values). There's a lot of interesting information in there, and it helps to know how to interpret it.

■ SAM file fields

These are the mandatory first 11 fields

Col	Field	Type	Brief description
1	QNAME	String	Query template NAME
2	FLAG	Int	bitwise FLAG
3	RNAME	String	References sequence NAME
4	POS	Int	1- based leftmost mapping POSition
5	MAPQ	Int	MAPping Quality
6	CIGAR	String	CIGAR string
7	RNEXT	String	Ref. name of the mate/next read
8	PNEXT	Int	Position of the mate/next read
9	TLEN	Int	observed Template LENgth
10	SEQ	String	segment SEQUENCE
11	QUAL	String	ASCII of Phred-scaled base QUALity+33

There can be many more fields and they also have valuable information.

Queen Felicity Rarely Plays Multi-dimensional Chess!

The "CIGAR" string (field #6 in the SAM output) tells you about the matches.

See https://en.wikipedia.org/wiki/List_of_mnemonics#Biology

And https://en.wikipedia.org/wiki/Sequence_alignment#Representations

CIGAR Code	BAM Integer	Description	Consumes query	Consumes reference
M	0	alignment match (can be a sequence match or mismatch)	yes	yes
I	1	insertion to the reference	yes	no
D	2	deletion from the reference	no	yes
N	3	skipped region from the reference	no	yes
S	4	soft clipping (clipped sequences present in SEQ)	yes	no
H	5	hard clipping (clipped sequences NOT present in SEQ)	no	no
P	6	padding (silent deletion from padded reference)	no	no
=	7	sequence match	yes	yes
X	8	sequence mismatch	yes	yes

The format is just concatenated repeats of COUNT CODE. For example: **1S22=1X13=2X30=3S**.

Install samtools

`samtools` can be used to work with SAM (and BAM - the small/fast binary format of SAM):

```
$ pixi add samtools
```

This gives you various useful commands:

```
$ samtools --help | less

$ samtools view -c matches-local-with-single.sam
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# Extract the FASTQ
$ samtools fastq matches-local-with-single.sam > matches.fastq
# Same thing, but compressed:
$ samtools fastq matches-local-with-single.sam | gzip > matches.fastq.gz

$ samtools flags

# Make a BAM file from a SAM
$ samtools view -b -h matches-local-with-single.sam > matches-local-with-single.bam
$ samtools sort matches-local-with-single.bam > matches-local-with-single-sorted.bam
$ samtools index matches-local-with-single-sorted.bam
```